

Financial Impact Analysis

To understand the real-world value of my Ensemble Model AUC 0.7850 and Threshold 0.48, here is a Scenario Assumption that translate statistical metrics into Currency (Euros):

Winning Model: Ensemble (XGBoost + LightGBM)

Performance: AUC 0.7850 | Threshold: 0.48

1. Scenario Parameters

Average Loan Amount: €10,000

Target Profit per "Good" Loan: €1,500

Net Loss per "Default": €7,000

2. Mathematical Appendix:

Baseline Profit (No Model Strategy): If the bank approved every applicant without a filter:

Total Good Clients: $56,538 (41,465 + 15,073) \times €1,500 = +€84,807,000$

Total Defaulters: $4,965 (3,449 + 1,516) \times €7,000 = -€34,755,000$

Total Baseline: €50,052,000

Baseline Profit per Application: $€50,052,000 / 61,503 = €813.81$

Ensemble Profit (Optimized Strategy): Using the 0.48 threshold to filter the portfolio:

Operational Profit (Approved Good): $41,465 \times €1,500 = +€62,197,500$

Risk Savings (Blocked Defaults): $3,449 \times €7,000 = +€24,143,000$

Operational Expenses (Missed Defaults): $1,516 \times €7,000 = -€10,612,000$

Opportunity Cost (Rejected Good): $15,073 \times €1,500 = -€22,609,500$

Total Ensemble: €53,119,000

Ensemble Profit per Application: $€53,119,000 / 61,503 = €863.70$

3. Final Portfolio Balance (Full 61,503 Applications)

Category	Impact	Value
Operational Profit	Profit from approved reliable clients	+ €62.20M

Risk Savings	Capital protected from potential defaults	+ €24.14M
Operational Expenses	Actual losses from missed defaults	- €10.61M
Opportunity Cost	Unrealized profit from rejections	- €22.61M
NET FINAL BALANCE		The Model's Total Economic Value + €53.12M

4. Ensemble Profit Strategic Conclusions:

Profitability:

The bank concludes the period with a positive net balance of €53.12 Million. The "Safety Tax": The €22.6M in profit "lost" by rejecting good clients is essentially an insurance premium the bank pays to avoid financial catastrophe. Without the model, the bank would have faced €34.75M in default losses (4,965×€7,000). With the model, those losses were slashed to just €10.61M.

Business Value:

$$€863.70 \text{ (Ensemble)} - €813.81 \text{ (Baseline)} = €49.89$$

The Efficiency Premium: Elevating Portfolio Value

Implementation of this advanced scoring system unlocks an estimated €53.12 Million in net profit per 60,000 applications. By masterfully balancing credit accessibility with capital preservation, the model elevates the average profit per applicant from €813.81 to €863.70. This represents a €49.89 "Efficiency Premium" generated by every single individual who enters our ecosystem. Even before a final decision is made (approved or not), the sheer precision of the model adds immediate economic value to the pipeline, ensuring that every application is an opportunity for optimized growth.

Risk Mitigation:

$$\text{Approvals: } 41,465 \text{ (True Negatives)} + 1,516 \text{ (False Negatives)} = 42,981 \text{ total approvals.}$$

$$\text{Default Rate: } (1,516 / 42,981) \times 100 = 3.53\% \text{ (down from 8.1\% baseline).}$$

The optimized model successfully identifies 70% of potential defaulters, reducing the portfolio's default rate from 8.1% to approximately 3.53%. Which means by applying the 0.48 threshold, the model have effectively cut the bank's risk exposure by more than half. It moved the portfolio from a high-risk baseline of 8.1% (where the bank approves everyone) to a much more stable and "bankable" 3.5%. This 4.6 percentage point drop represents the €24.14M in risk savings calculated in the financial balance report.

Thank you.